PRML COURSE PROJECT

Group 8

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PROBLEM STATEMENT

Visual Question Answering

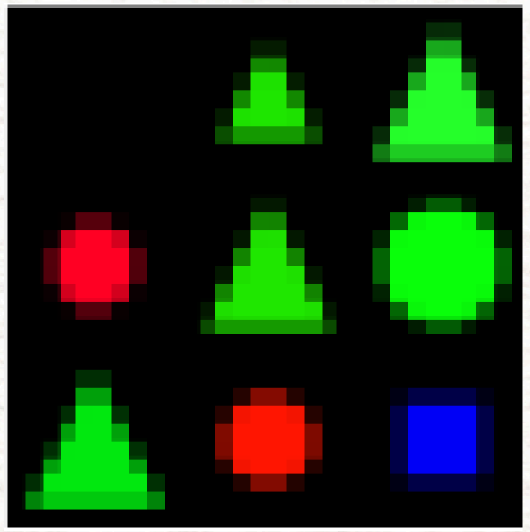
Given an image and a question, the model must generate an answer by understanding object attributes (e.g., color, shape), object types, and spatial/logical relationships between objects.

Input

Image + Query

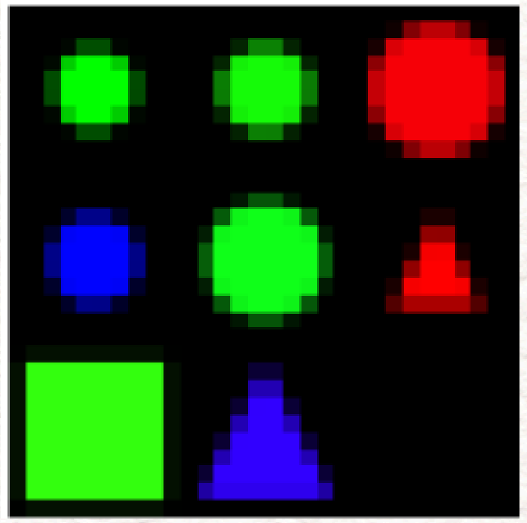
Output

True / False



(is triangle (above circle))

YES/NO



(is red (right_of (left_of circle)))

YES/NO

DATASET DESCRIPTION

WHAT WE ARE GIVEN

We were provided with the SHAPES dataset, in which the data is divided into train(large, medium, small, tiny), validation and test data.

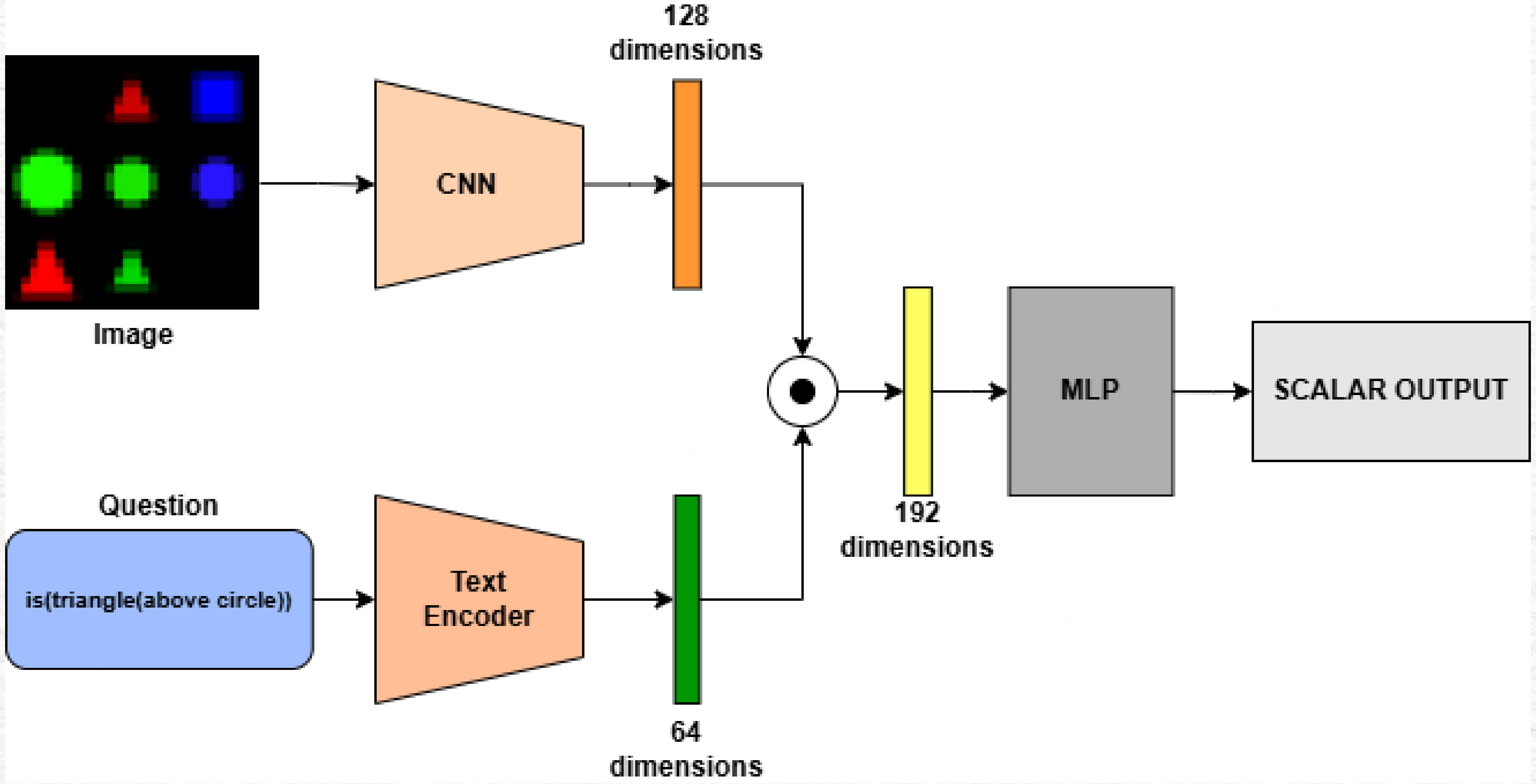
Each sample contains an image (of dimension 30 x 30 x 3), a query, and a ground truth binary label.

OBSERVATION

- Images contain simple geometric objects:
    a. Shapes: circle, square, triangle
    b. Colours: red, green, blue
- Objects are arranged in a grid like layout, where each cell of size 10X10 represents a shape, thus making the size of complete image as 30 x 30 x 3 (3 color channels)

MODEL USED

APPROACH-1:



CNN

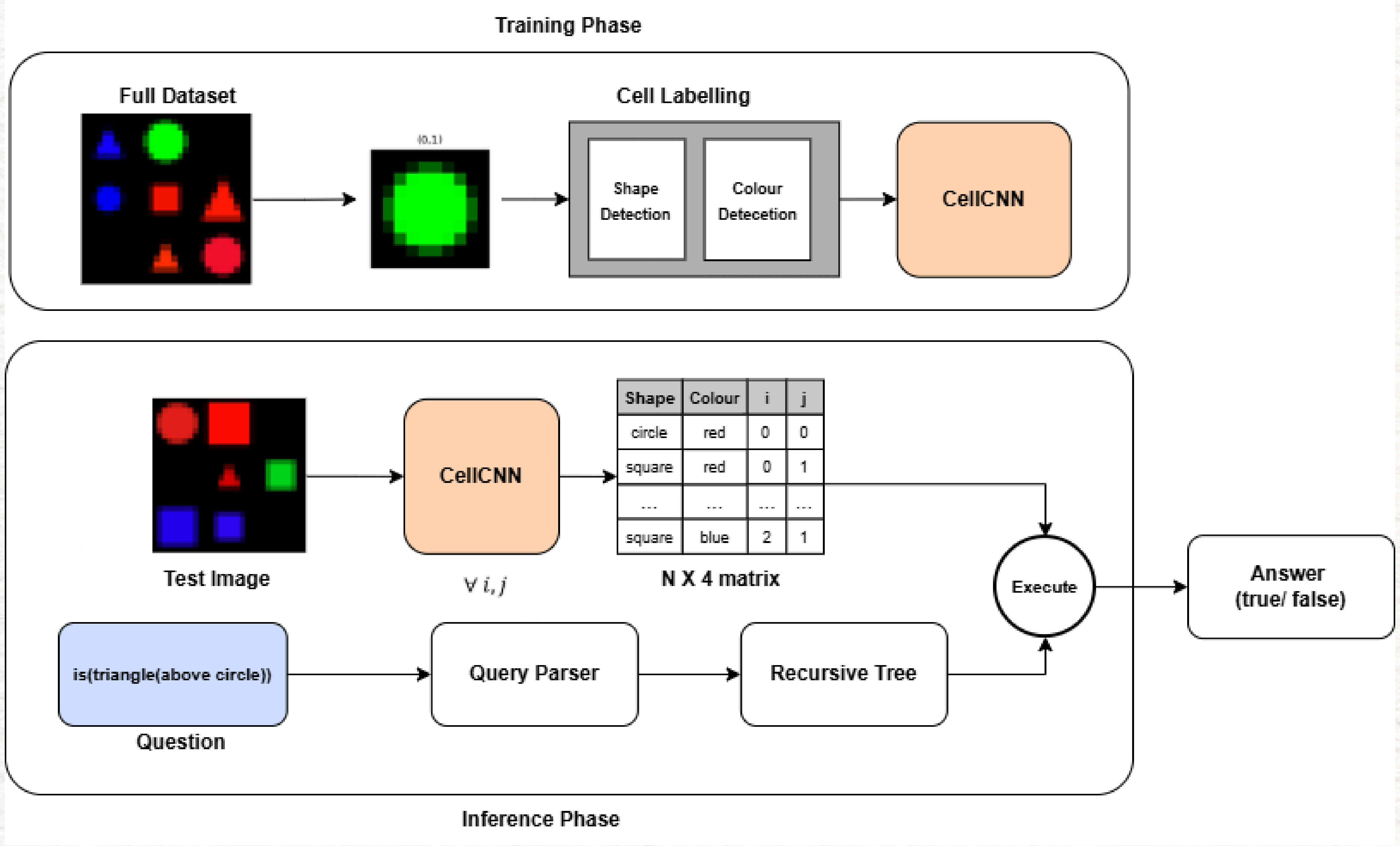
+

MEAN EMBEDDING

(LATE FUSION)

MODEL USED

APPROACH-2:



CNN

+

RULE BASED QUERY PARSING

EFFORTS AND METHODOLOGY

APPROACH-1: CNN + EMBEDDING

Image Encoder: 3-layer CNN (Conv, BatchNorm, ReLU)→ 128-d feature vector
Text Encoder: Token embeddings + mean pooling → 64-d vector
Fusion: Concatenation of image and text features
Classifier: 2-layer MLP → Scalar output

LIMITATION:

- **No compositional structure**: Queries with identical tokens but different nesting (e.g., “(below (left_of square))” vs “(left_of (below square))”) are treated similarly, despite different meanings.
- **Order & spatial info lost**: Averaging embeddings ignores word order (e.g., “(left_of red blue)” vs “(left_of blue red)”), and global pooling (AdapAvgPool12d) removes position-object relationships.
- **Cannot handle recursion**: Fixed-depth MLP cannot process arbitrarily nested queries, so it fails on deeper compositions.

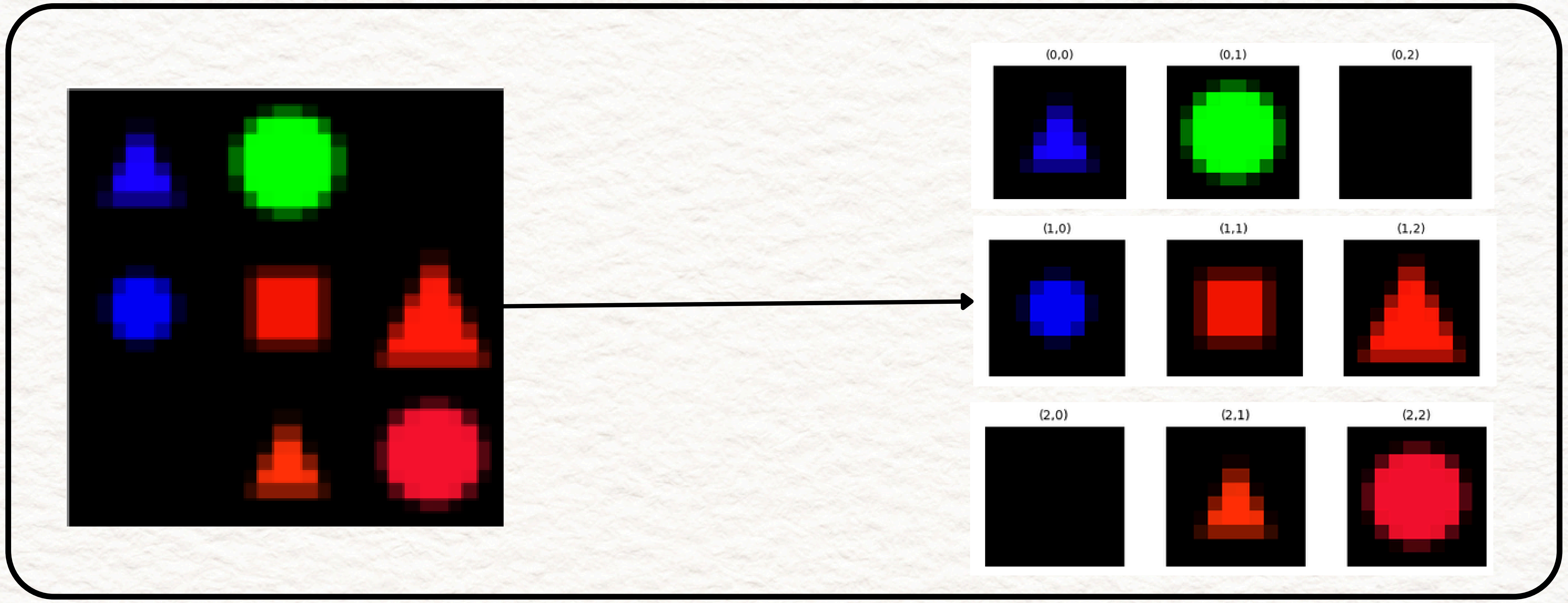
APPROACH-2: CNN + RULE-BASED QUERY PARSING

Since the dataset demands us to capture the relative information between objects, we used CNN perception + Rule-based query parsing, which performs deep learning on the vision task (detecting the shape and its color) using CNN, and a recursive logic for parsing the query.

STAGE - 1: VISUAL PERCEPTION (CNN)

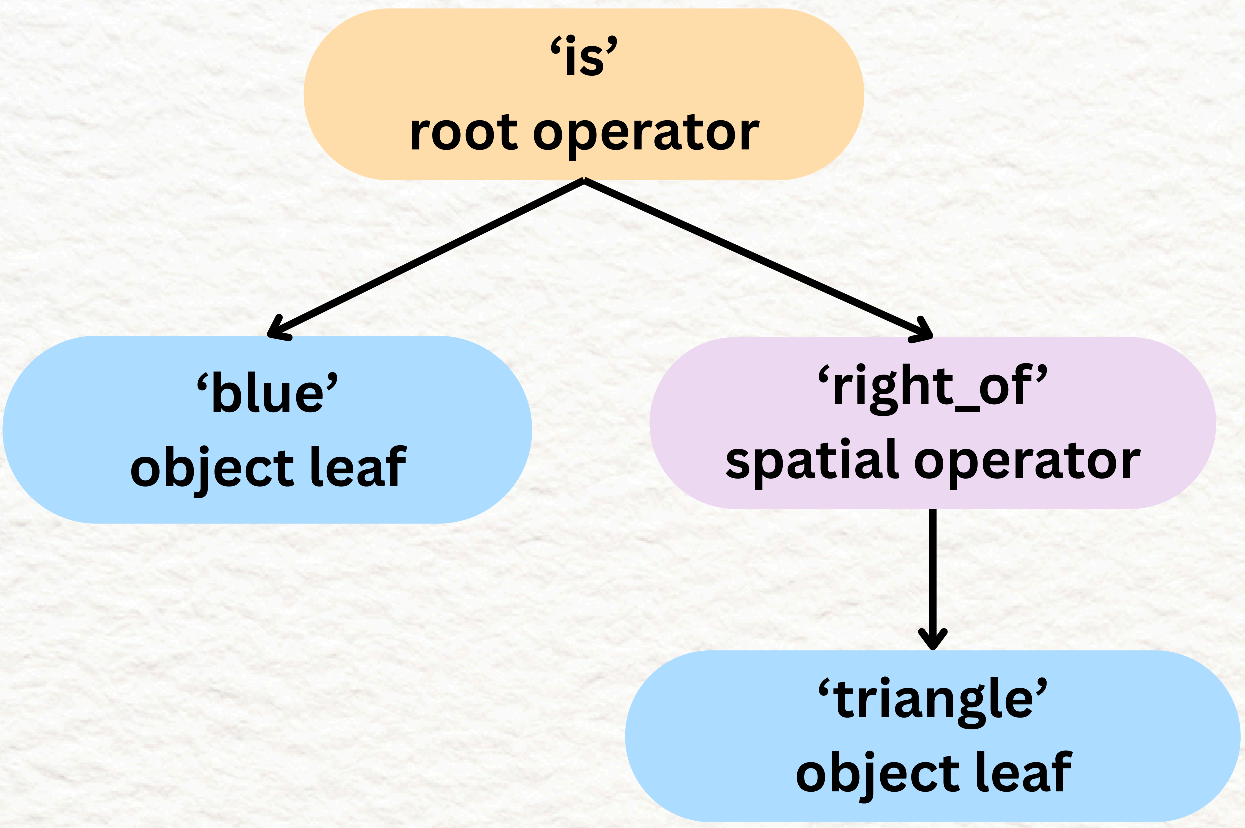
- To pave the way for relational reasoning during query parsing, instead of training the CNN on the provided dataset we divide the 3x3 grid image (30x30) into 9 cells containing one object each (10x10) and train the CNN to detect color and shape given cells like these as input.
- However, we do not have the corresponding labels to train the CNN on these cells.
- We use heuristics based on geometric patterns to approximately assign labels to these cells. Although this does not guarantee complete correctness of the labels, it is sufficient for a simple CNN like ours, especially when trained on sufficient amount of cells over all splits.

- Now, at inference time, the CNN is applied on all 9 cells of the given image, and the information about the detected shapes and their positions are stored in an array.



STAGE - 2: QUERY PARSING

- It was observed from the dataset that the questions were formatted as parenthesized prefix logic queries (e.g., (is red (left_of circle)).
- Hence, we parse the query in a way which generates a recursive tree which will be used to apply operations based on words like “above”, “right_of”, etc.



(is blue (right_of triangle))

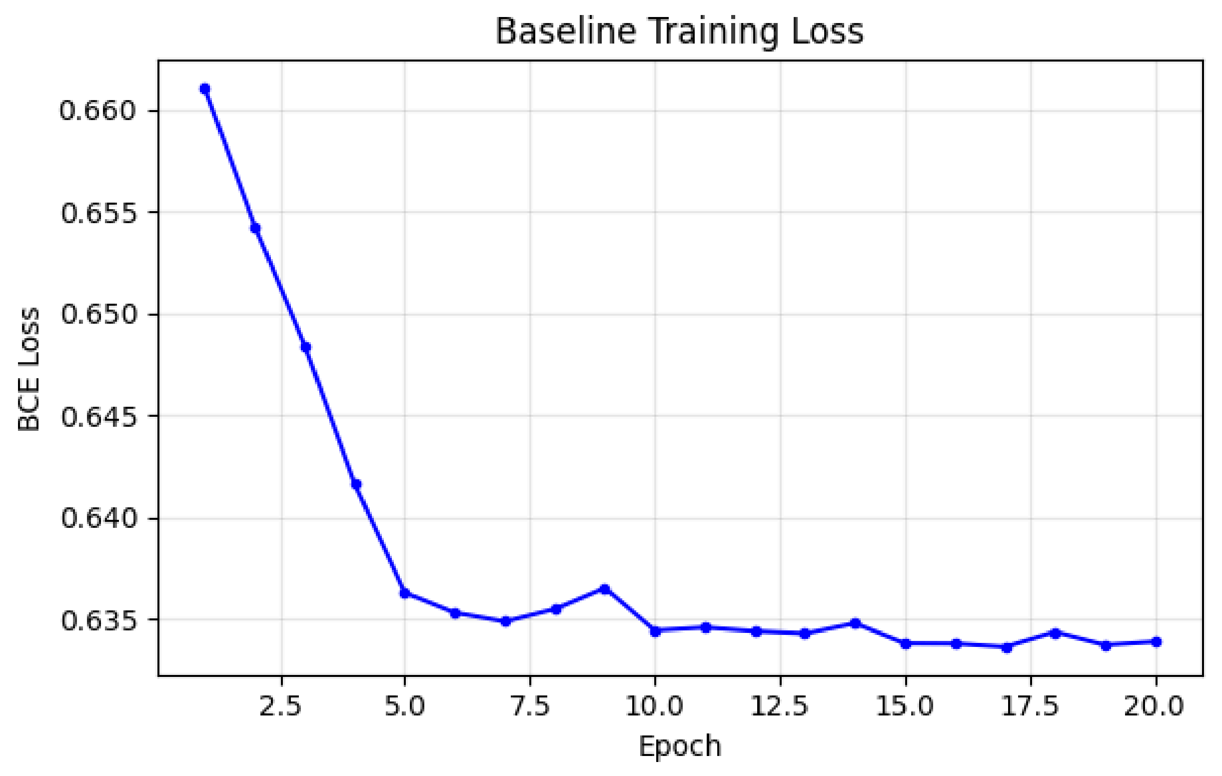
STAGE- 3: COMPOSITIONAL REASONING

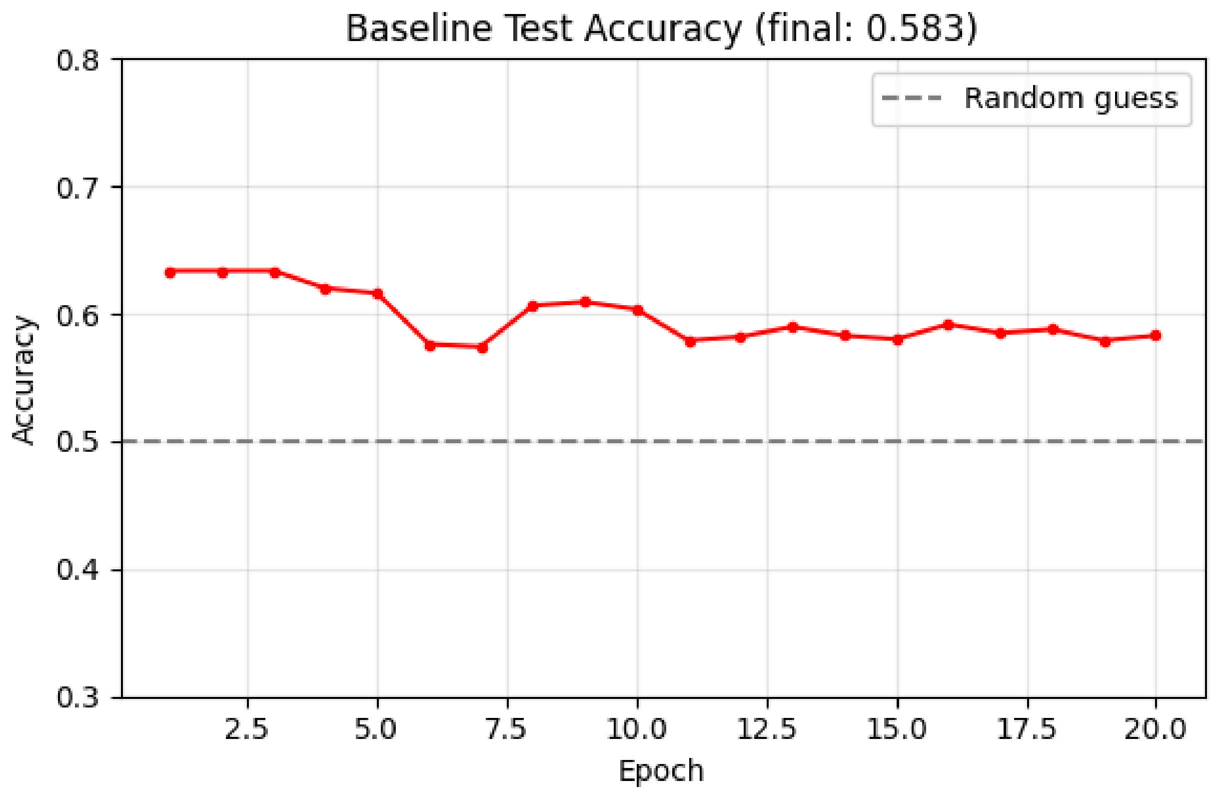
- After the image has been turned into a list of objects (with coordinates, shapes and colours) by the CNN, and the query parser has turned the text into a tree, the execute function combines them.
- It recursively evaluates the tree:
  - **Spatial Rules**: It uses basic coordinate math to execute operators like left_of, right_of, above, and below based on the grid index of the detected shapes.
  - **Logical Operators**: It filters sets of objects based on properties (e.g., the is operator checks if an object matches a requested color or shape).

RESULTS AND ANALYSIS

APPROACH-1: CNN + EMBEDDING (BASELINE APPROACH)

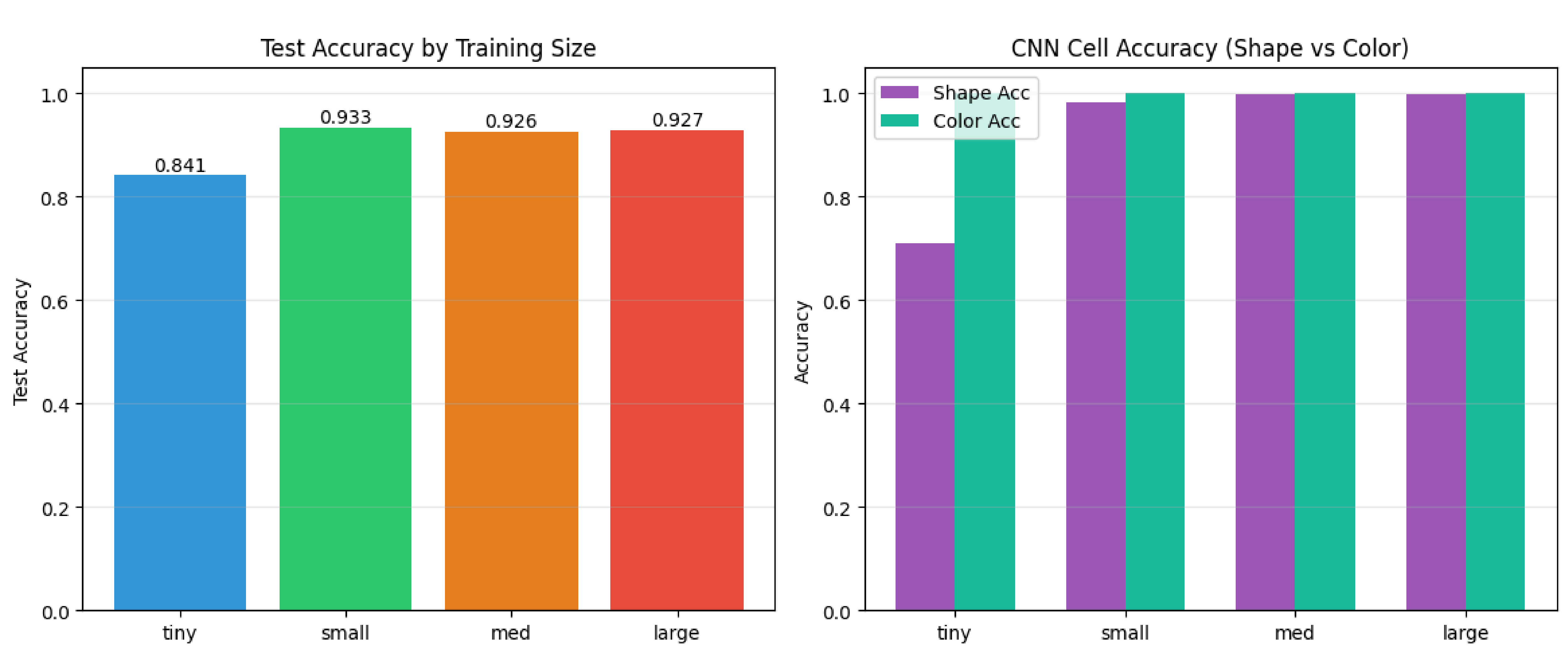
- The model achieved a final test accuracy of ~58.30%, barely outperforming a simple 50% random guess baseline.
- Despite 20 epochs of training, the BCE Loss plateaued early (around Epoch 8), indicating the model was unable to extract deeper hierarchical features from the fused data.
- The model showed high variance in accuracy across epochs, suggesting it was "guessing" based on statistical noise rather than learning the underlying spatial logic.





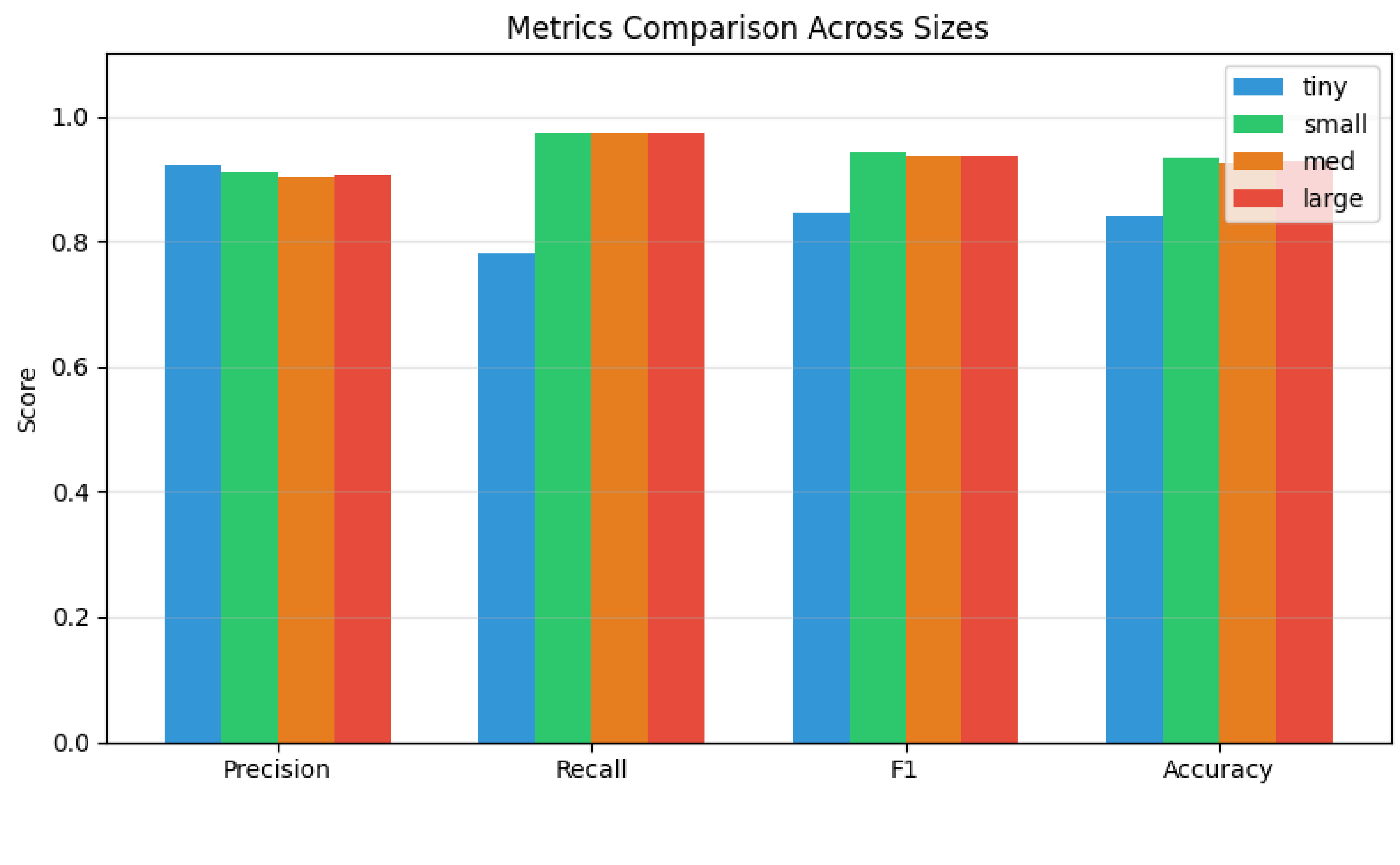
APPROACH-2: CNN + RULE-BASED QUERY PARSING

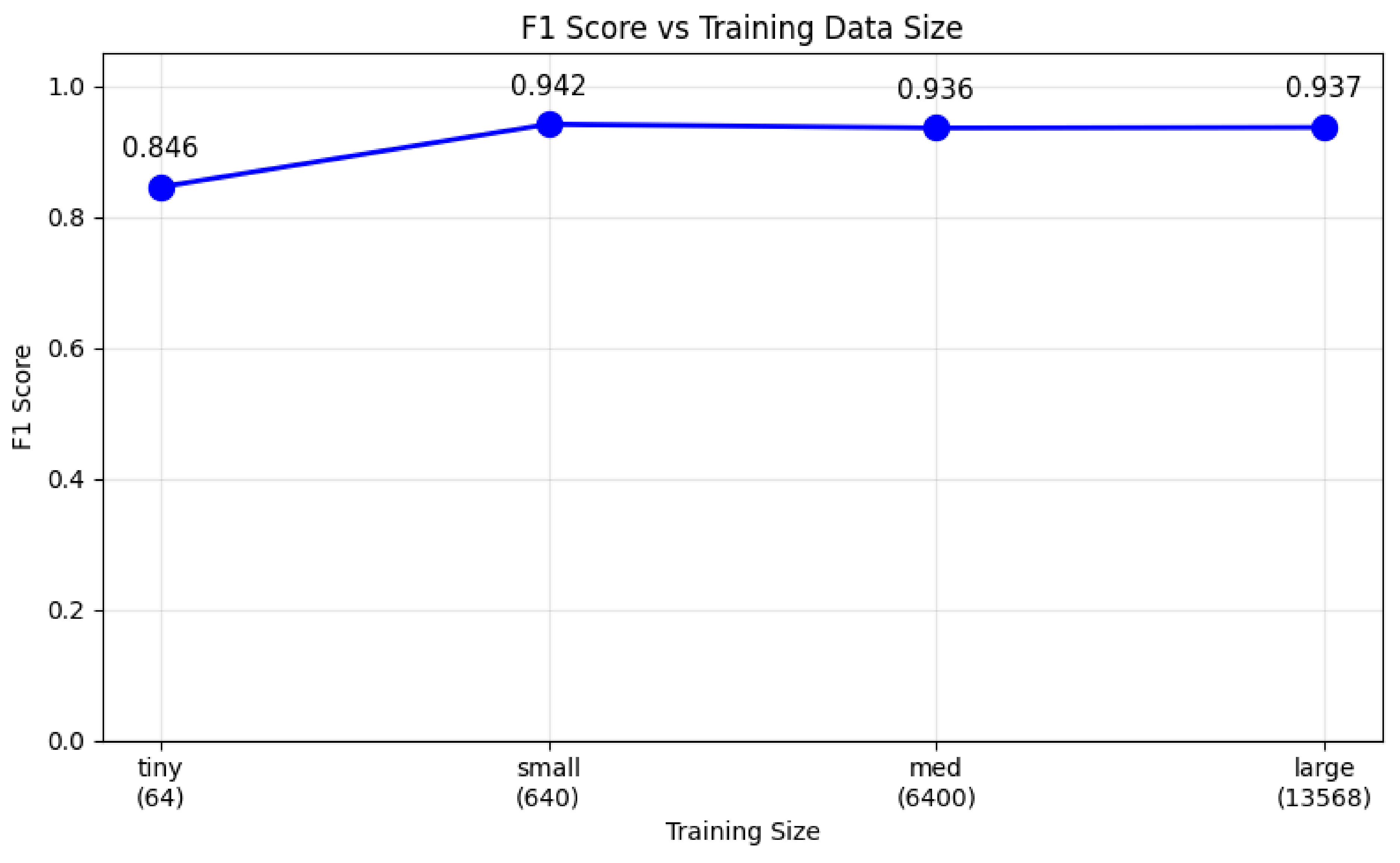
- The model achieved a test accuracy of ~93% when trained on small, medium and large datasets, and an accuracy of 84.1 % on the tiny dataset.
- The graphs show the comparison of test accuracy and perception(shape/colour) performance scaling relative to training dataset size.



APPROACH-2: CNN + RULE-BASED QUERY PARSING

- Comparison of different performance metrics - Precision, Recall, F1 Score and Accuracy across various training dataset sizes
- We see that while better precision is obtained on the tiny dataset, the other performance metrics are highest on the small dataset.

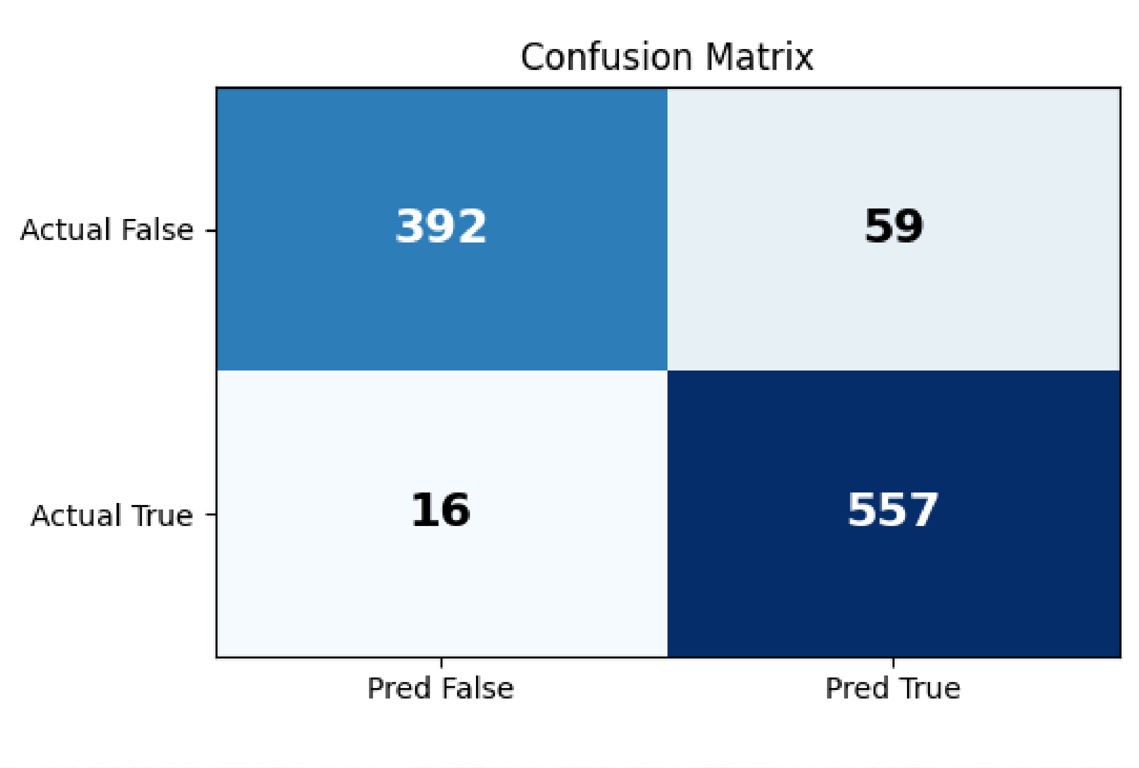


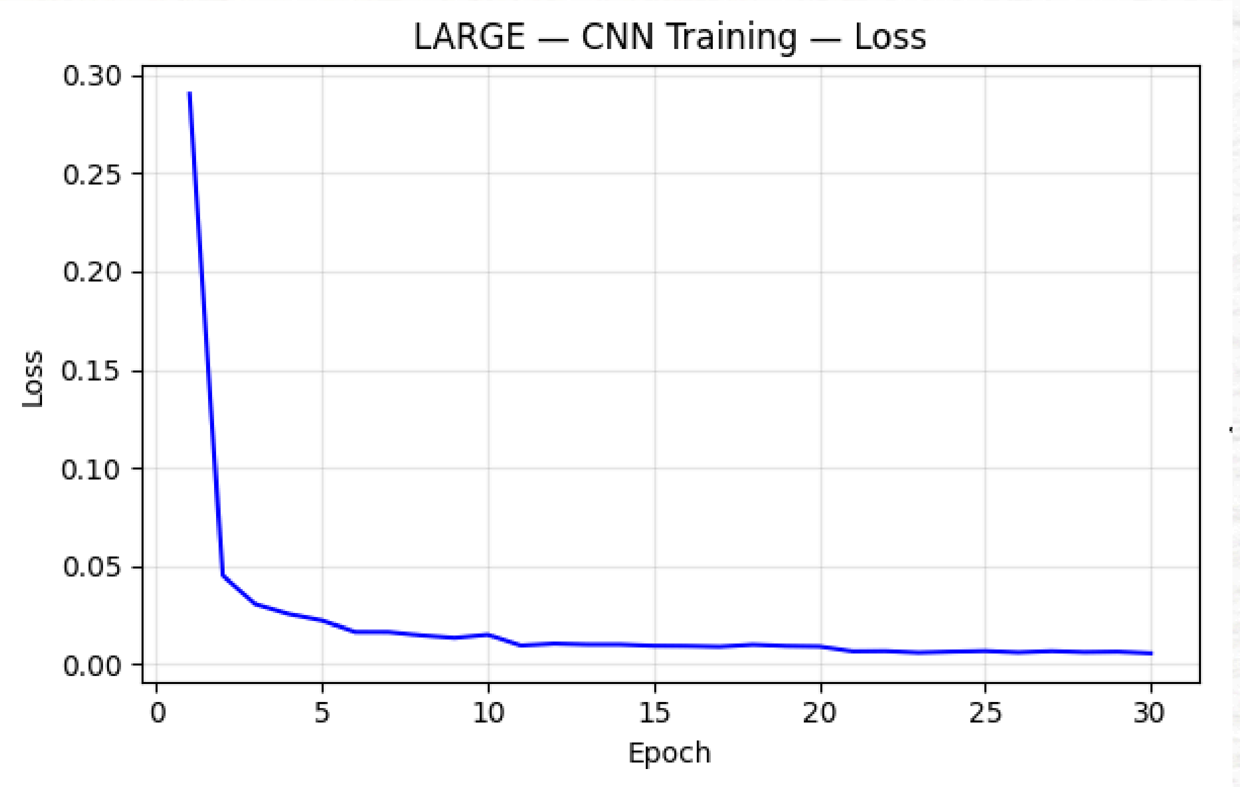


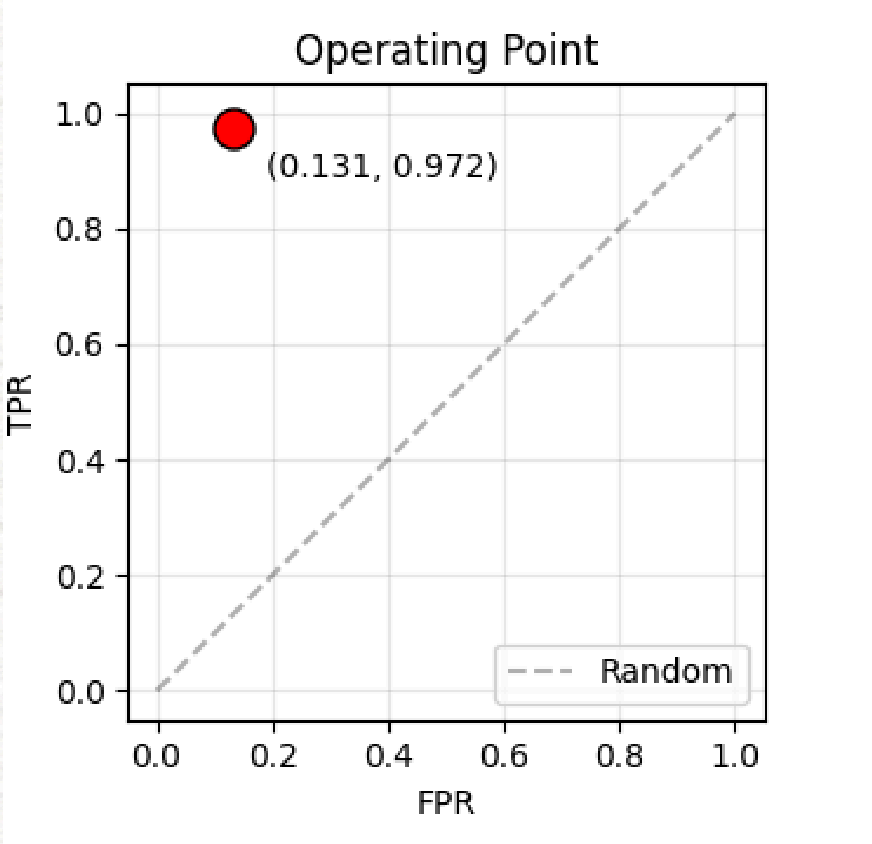
- Detailed performance metrics when trained on the large dataset:

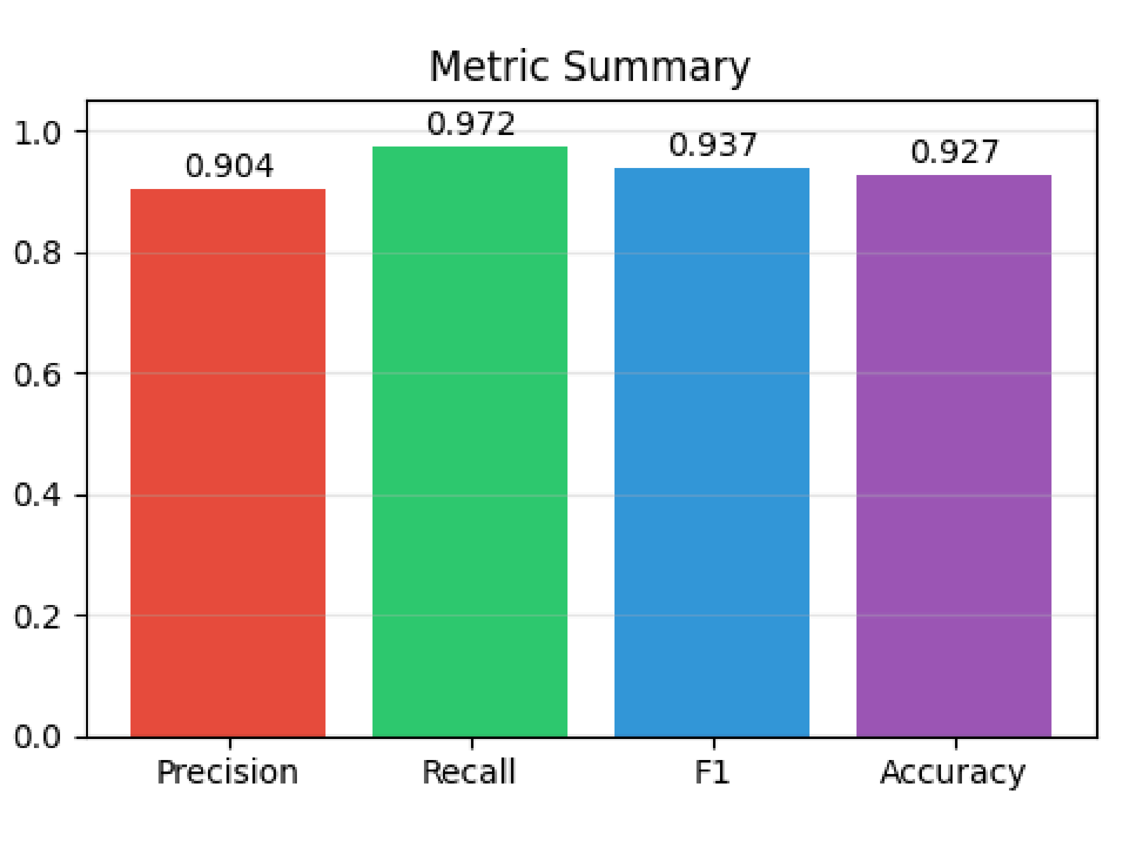
Accuracy: 0.9268
Precision: 0.9042
Recall: 0.9721
F1 Score: 0.9369

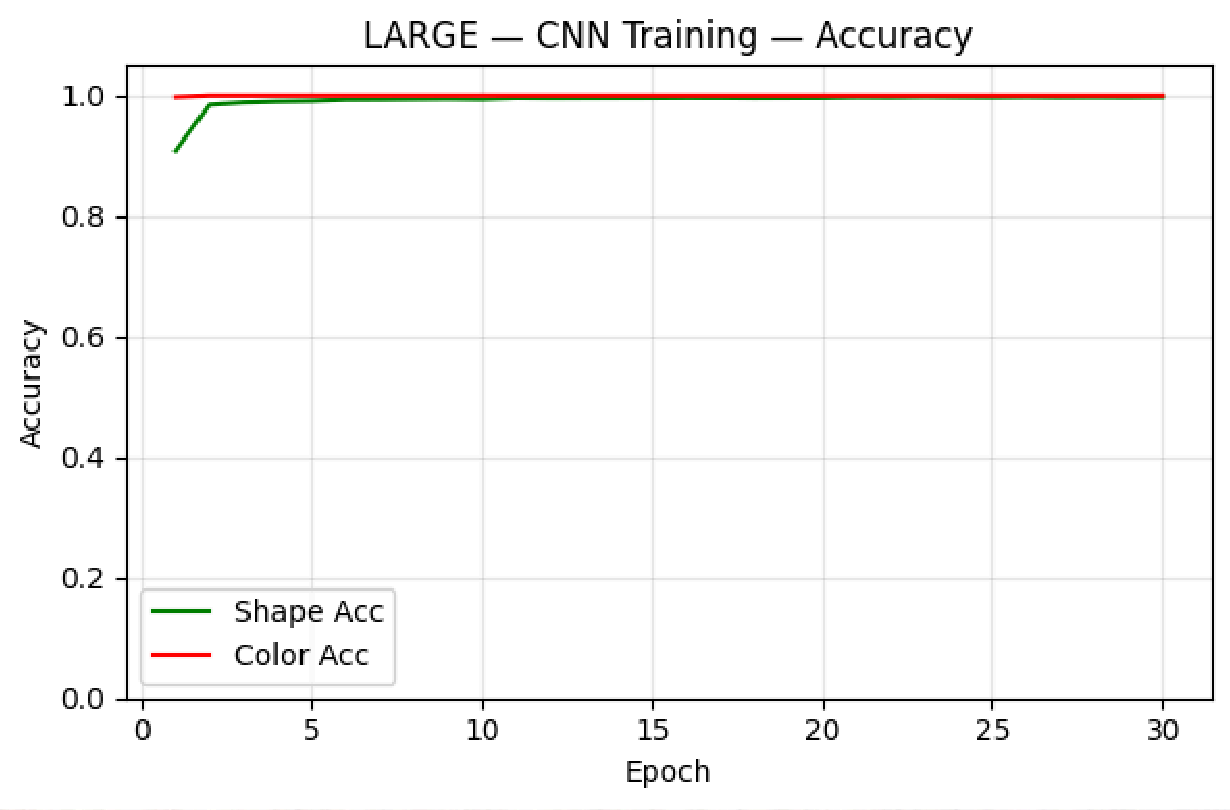
Confusion:
TP=557 FP=59
FN=16 TN=392











WORK DIVISION

Anarghya Das: Cell Classifier CNN, Heuristic labelling, MLP fusion
Raghav Srinivasan: Parser, Executor, Analysis
Subhanshu Gupta: Cell Classifier CNN, Executor, Text encoder
Saksham Panghal: Parser, CNN image encoder, Analysis

THANK YOU